

AWS Cloud Practical

Name: Pooja Prakash Belgaumkar

1. Create a NAT gateway and test internet access from a private EC2

- **Public subnet** → has Internet Gateway (IGW)
 - **NAT Gateway** → sits in public subnet
 - **Private subnet** → routes internet traffic to NAT
 - **Private EC2** → uses NAT to go out, but cannot be accessed from internet
- Create a custom VPC named **nat-vpc** with an appropriate IPv4 CIDR block (for example, 10.0.0.0/16).

The screenshot shows the AWS Management Console page for a VPC named **nat-vpc** (ID: vpc-0a749342fb4666100). The page displays various configuration details in a grid format:

Details Info			
VPC ID vpc-0a749342fb4666100	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0412432b18ea86b4c	Main route table rtb-0b7061b5f038093a7
Main network ACL acl-0b5f8c86b6f108351	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 787169320097
Encryption control ID -	Encryption control mode -		

- Create an Internet Gateway and attach it to the **nat-vpc**. This will allow resources in public subnets to access the internet.

The screenshot shows the AWS Management Console page for an Internet Gateway named **nat-igw** (ID: igw-0a1b420bf7a0e4071). A green notification bar at the top states: "Internet gateway igw-0a1b420bf7a0e4071 successfully attached to vpc-0a749342fb4666100". The details section shows:

Details Info			
Internet gateway ID igw-0a1b420bf7a0e4071	State Attached	VPC ID vpc-0a749342fb4666100 nat-vpc	Owner 787169320097

Below the details, the **Tags (1)** section shows a single tag:

Key	Value
Name	nat-igw

- Create one public subnet and one private subnet inside the nat-vpc.
- Enable Auto-assign Public IPv4 Address for the public subnet only. Do not enable public IP assignment for the private subnet.

Subnets (2) [Info](#) Last updated 1 minute ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

VPC: vpc-0a749342fb4666100 [Clear filters](#)

☐	Name	Subnet ID	State	VPC	Block Public...	IPv
☐	nat-public-subnet	subnet-018319b7d717e1d84	Available	vpc-0a749342fb4666100 nat-vpc	Off	10
☐	nat-private-subnet	subnet-03ff904849543c73b	Available	vpc-0a749342fb4666100 nat-vpc	Off	10

- Create a route table for the public subnet.
- Add a route with:
 - **Destination:** 0.0.0.0/0
 - **Target:** Internet Gateway (IGW)
- Associate this route table with the public subnet.

[VPC](#) > [Route tables](#) > rtb-0eac5d5de96930d11 Info [Actions](#)

rtb-0eac5d5de96930d11 / nat-public-rt

Details [Info](#)

Route table ID
[rtb-0eac5d5de96930d11](#)

VPC
[vpc-0a749342fb4666100](#) | nat-vpc

Main
[No](#)

Owner ID
[787169320097](#)

Explicit subnet associations
[subnet-018319b7d717e1d84](#) / [nat-public-subnet](#)

Edge associations
-

[Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Routes (2) [Both](#) [Edit routes](#)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0a1b420bf7a0e4071	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

- Create a separate route table for the private subnet. At this stage, do not add any internet route and do not associate it yet.

☰ [VPC](#) > [Route tables](#) > rtb-054fc0d73a8edaa94 ⓘ 🖨️ 🔔

rtb-054fc0d73a8edaa94 / nat-private-rt Actions ▾

Details [Info](#)

Route table ID
 rtb-054fc0d73a8edaa94

VPC
 vpc-0a749342fb4666100 | [nat-vpc](#)

Main
 No

Owner ID
 787169320097

Explicit subnet associations
 -

Edge associations
 -

[Routes](#) | [Subnet associations](#) | [Edge associations](#) | [Route propagation](#) | [Tags](#)

Routes (1) Both ▾ Edit routes

< 1 > ⚙️

Destination ▾	Target ▾	Status ▾	Propagated ▾	Route Origin ▾
10.0.0.0/16	local	🟢 Active	No	Create Route Table

- Create a NAT Gateway inside the public subnet. Create a NAT Gateway inside the public subnet.
- Wait until the NAT Gateway status changes to Available.

☰ [VPC](#) > [NAT gateways](#) > nat-1393d02109b7ace75 ⓘ 🖨️ 🔔

nat-1393d02109b7ace75 / my-nat-gateway Actions ▾

Details

NAT gateway ID
 nat-1393d02109b7ace75

NAT gateway ARN
 arn:aws:ec2:eu-north-1:787169320097:natgateway/nat-1393d02109b7ace75

VPC
 vpc-0a749342fb4666100 / [nat-vpc](#)

Availability mode
 Regional

Connectivity type
 Public

Method of EIP allocation
 Automatic

State
 Available

Created
 Tuesday, February 10, 2026 at 00:59:25 GMT+5:30

State message [Info](#)
 -

Deleted
 -

[IP addresses](#) | [Monitoring](#) | [Flow logs](#) | [Tags](#)

Associated IP addresses (1) 🔄 Edit IP address associations

< 1 > ⚙️

IP address ▾	Status ▾	Availability Zone ▾	Allocation ID ▾
13.62.81.155	🟢 Succeeded	eun1-az3 (eu-north-1c)	eipalloc-00834bf3133f

13.62.81.155

Actions ▾

Associate Elastic IP address

Summary

Allocated IPv4 address

13.62.81.155

Association ID

eipassoc-08dd4b6b7703378fb

Network interface ID

-

Address pool

Amazon

Type

Public IP

Scope

VPC

Network interface owner account ID

-

Network border group

eu-north-1

Allocation ID

eipalloc-00834bf3133fa8330

Associated instance ID

-

Public DNS

-

Service managed

rnat

Reverse DNS record

-

Private IP address

13.62.81.155

NAT Gateway ID

nat-1393d02109b7ace75 (my-nat-gateway)

- Then after creating the NAT Gateway update the Private Route Table route:
 - Destination: 0.0.0.0/0
 - Target: NAT Gateway
- This allows private subnet instances to access the internet through the NAT Gateway.

Edit routes

Route 1

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Route 2

Destination

0.0.0.0/0

Target

NAT Gateway

Status

-

Propagated

No

nat-1393d02109b7ace75

Use: "nat-1393d02109b7ace75"

nat-1393d02109b7ace75 (my-nat-gateway)

Remove

Add route

Cancel

Preview

Save changes

- Associate the updated private route table with the **private subnet**.

VPC > Route tables > rtb-054fc0d73a8edaa94 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	nat-public-subnet	subnet-018319b7d717...	10.0.1.0/24	-	rtb-0eac5d5de96930d11 / nat-p...
☒	nat-private-subnet	subnet-03ff904849543...	10.0.2.0/24	-	Main (rtb-0b7061b5f038093a7)

Selected subnets

subnet-03ff904849543c73b / nat-private-subnet

Cancel Save associations

- Then create the Security Groups for both Public EC2 and Private EC2.
- In Public Security group edit inbound rule to SSH(22) → MY IP

VPC > Security Groups > sg-0f7bcc7678fd3225f - nat-public-sg

sg-0f7bcc7678fd3225f - nat-public-sg

Details

Security group name nat-public-sg	Security group ID sg-0f7bcc7678fd3225f	Description Allows SSH access to developers	VPC ID vpc-0a749342fb4666100
Owner 787169320097	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing VPC associations Related resources - new Tags

Inbound rules (1)

Search

Security group rule ID	IP version	Type	Protocol	Port range	Source
sg-0c47143c1a9ee14ad	IPv4	SSH	TCP	22	106.212.187.68/32

- In private Security Group edit inbound rule SSH(22) → Source = public SG, because Only the public EC2 (bastion) can SSH into the private EC2

VPC > Security Groups > sg-0cef2d623acc018d4 - nat-private-sg

sg-0cef2d623acc018d4 - nat-private-sg

Details

Security group name nat-private-sg	Security group ID sg-0cef2d623acc018d4	Description Allows SSH access to developers	VPC ID vpc-0a749342fb4666100
Owner 787169320097	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing VPC associations Related resources - new Tags

Inbound rules (1)

Search

Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
sg-0d55b76d270febd8e	-	SSH	TCP	22	sg-0f7bcc7678fd3225f / nat-public-sg	-

- Launch a **Public EC2 instance** in the public subnet and attach the public security group.

EC2 > Instances > i-07d2c5a9f4a26b381

Instance summary for i-07d2c5a9f4a26b381 (nat-public-ec2) [Info](#)

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-07d2c5a9f4a26b381	Public IPv4 address 13.53.214.222 open address	Private IPv4 addresses 10.0.1.115
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-1-115.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-1-115.eu-north-1.compute.internal	
Answer private resource DNS name -	Instance type t3.micro	Elastic IP addresses -
Auto-assigned IP address 13.53.214.222 [Public IP]	VPC ID vpc-0a749342fb4666100 (nat-vpc)	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
IAM Role -	Subnet ID subnet-018319b7d717e1d84 (nat-public-subnet)	Auto Scaling Group name -
IMDSv2 Required	Instance ARN arn:aws:ec2:eu-north-1:787169320097:instance/i-07d2c5a9f4a26b381	Managed false
Operator		

- Launch a **Private EC2 instance** in the private subnet and attach the private security group.
- Use the same key pair for both instances.

EC2 > Instances > i-0f22c045c68fcd4d

Instance summary for i-0f22c045c68fcd4d (nat-private-instance) [Info](#)

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-0f22c045c68fcd4d	Public IPv4 address -	Private IPv4 addresses 10.0.2.220
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-2-220.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-2-220.eu-north-1.compute.internal	
Answer private resource DNS name -	Instance type t3.micro	Elastic IP addresses -
Auto-assigned IP address -	VPC ID vpc-0a749342fb4666100 (nat-vpc)	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
IAM Role -	Subnet ID subnet-03ff904849543c73b (nat-private-subnet)	Auto Scaling Group name -
IMDSv2 Required	Instance ARN arn:aws:ec2:eu-north-1:787169320097:instance/i-0f22c045c68fcd4d	Managed false

- SSH into the public EC2 from your local machine using the key pair.

```
PS C:\Users\P00JA\Downloads> ssh -i C:\Users\P00JA\Downloads\linux1.pem ubuntu@13.53.214.222
The authenticity of host '13.53.214.222 (13.53.214.222)' can't be established.
ED25519 key fingerprint is SHA256:dEir5CAz4eFKAuUI+Pg01r8K/kZDkChY99pSl/Hvngg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.53.214.222' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Feb  9 20:11:43 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   25.9% of 6.71GB Processes:      111
Memory usage: 23%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.1.115

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-0-1-115:~$ |
```

- Exit to your local machine and copy the key pair to the public EC2 using SCP:

scp -i linux1.pem linux1.pem ubuntu@<Public-EC2-Public-IP>:/home/ubuntu/

- This step is required to allow SSH access from the public EC2 to the private EC2.

```
ubuntu@ip-10-0-1-115:~$ exit
logout
Connection to 13.53.214.222 closed.
PS C:\Users\P00JA\Downloads> scp -i linux1.pem linux1.pem ubuntu@13.53.214.222:/home/ubuntu/
linux1.pem                               100% 1678      6.5KB/s   00:00
PS C:\Users\P00JA\Downloads> ssh -i linux1.pem ubuntu@13.53.214.222
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)
```

- Then SSH back into the public EC2 and secure the key and then try to connect it to the Private EC2

```
ubuntu@ip-10-0-1-115:~$ chmod 400 linux1.pem
ubuntu@ip-10-0-1-115:~$ ssh -i linux1.pem ubuntu@10.0.2.220
The authenticity of host '10.0.2.220 (10.0.2.220)' can't be established.
ED25519 key fingerprint is SHA256:gHONA1ATMhBP3Y+dgYlk8wdpdds79DTmukGEt126QCg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.220' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Feb  9 20:22:16 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   26.1% of 6.71GB Processes:    111
Memory usage: 24%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.2.220

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

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individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-0-2-220:~$ |
```

- Test Internet Access from the Private EC2
- Successful output confirms that the NAT Gateway is working correctly.

```
ubuntu@ip-10-0-2-220:~$ ping google.com
PING google.com (142.251.142.238) 56(84) bytes of data.
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=1 ttl=118 time=4.05 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=2 ttl=118 time=3.82 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=3 ttl=118 time=3.85 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=4 ttl=118 time=3.84 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=5 ttl=118 time=3.85 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=6 ttl=118 time=3.85 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=7 ttl=118 time=3.85 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=8 ttl=118 time=3.84 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=9 ttl=118 time=3.84 ms
^C
--- google.com ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8015ms
rtt min/avg/max/mdev = 3.816/3.864/4.048/0.065 ms
ubuntu@ip-10-0-2-220:~$
```